

■ NETWORKING EXAM COMPLETE CHEAT SHEET

Covers: Static NAT • NAT Types • NTP • DHCP • OSPF • RIP • Static Routing — All commands included

1. NAT — Network Address Translation

1.1 Key Concepts

Inside Local – Private IP inside the org network | **Inside Global** – How inside IP appears on outside

Outside Local – How outside IP appears inside | **Outside Global** – Real outside IP

1.2 NAT Types

Type	Description
Static NAT	1-to-1 mapping. One inside IP → one outside IP. Permanent.
Dynamic NAT	Pool of outside IPs. Inside IPs mapped from pool. If pool exhausted → dropped.
PAT / Overload	Many inside IPs share ONE outside IP using different port numbers. Most common (home routers).

1.3 Private IP Address Ranges

Class	Start IP	End IP	# Hosts
A	10.0.0.0	10.255.255.255	16,777,216
B	172.16.0.0	172.31.255.255	1,048,576
C	192.168.0.0	192.168.255.255	65,536

1.4 Static NAT Configuration — Step by Step

Step	Command	Purpose
1	router# configure terminal	Enter global config mode
2	router(config)# ip nat inside source static <local-ip> <global-ip>	Map private IP to public IP (repeat for each host)
3	router(config)# interface <inside-interface>	Enter inside interface config (e.g. fa0/0)
4	router(config-if)# ip nat inside	Mark interface as NAT inside
5	router(config-if)# interface <outside-interface>	Enter outside interface config (e.g. se0/0/0)
6	router(config-if)# ip nat outside	Mark interface as NAT outside
7	router(config-if)# end	Exit config mode

1.5 Static NAT Full Example (from lab)

```
! Router0 (r1) – public side
r1(config)# int fa0/0
r1(config-if)# ip add 1.0.0.1 255.0.0.0 | no shut
r1(config)# int se0/0/0 | clock rate 64000 | ip add 2.0.0.1 255.0.0.0
r1(config)# ip route 3.0.0.0 255.0.0.0 2.0.0.2

! Router1 (r2) – NAT router (private side)
r2(config)# int se0/0/0 | ip add 2.0.0.2 255.0.0.0 | no shut
r2(config)# int fa0/0 | ip add 10.0.0.1 255.0.0.0 | no shut
r2(config)# ip route 1.0.0.0 255.0.0.0 2.0.0.1
r2(config)# ip nat inside source static 10.0.0.2 3.0.0.2
r2(config)# ip nat inside source static 10.0.0.3 3.0.0.3
r2(config)# ip nat inside source static 10.0.0.4 3.0.0.4
r2(config)# ip nat inside source static 10.0.0.5 3.0.0.5
r2(config)# int fa0/0 | ip nat inside
r2(config)# int se0/0/0 | ip nat outside
```

1.6 NAT Verification Commands

Command	Description
r2# show ip nat translations	Shows active NAT translation table
r2# show ip nat statistics	Shows NAT stats (hits, misses, etc.)
r2# debug ip nat	Real-time NAT translation debug output
r2# clear ip nat translation *	Clears all dynamic NAT translations

■ If ping says 'Destination host unreachable' — NAT is not translating. Check ip nat inside/outside on correct interfaces!

2. NTP — Network Time Protocol

NTP synchronizes clocks on network devices. Cisco routers have: (1) Hardware clock = 'calendar', (2) Software clock = 'clock' (primary). Services like syslog depend on accurate time.

2.1 NTP Configuration

```
! Step 1: Configure IP on router
r1(config)# int fa0/0 | ip add 1.0.0.1 255.0.0.0 | no shut

! Step 2: Check time BEFORE NTP
r1# show clock → shows wrong time e.g. *0:8:25.144 UTC Mon Mar 1 1993

! Step 3: On NTP SERVER – enable NTP in Server0 > Services > NTP > ON

! Step 4: Configure router as NTP CLIENT
r1(config)# ntp server 1.0.0.2 ← IP of NTP server

! Step 5: Verify
r1# show clock → should show correct time now
r1# show ntp status → shows 'Clock is synchronized, stratum 2'
```

3. DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns IP addresses, subnet mask, default gateway, DNS to clients. Very common in home & enterprise networks.

3.1 DHCP 4-Step Process (DORA)

Step	Message	Type	Description
1	DHCPDISCOVER	Broadcast → 255.255.255.255	Client broadcasts: 'Is there a DHCP server?'
2	DHCPOFFER	Unicast to MAC	Server offers: IP, subnet, gateway, DNS, lease time
3	DHCPREQUEST	Broadcast	Client accepts the first offer received
4	DHCPACK	Unicast	Server confirms the lease. Also used to renew lease.

3.2 Configure Router as DHCP Server

```
! Exclude IP addresses (reserved for servers/static devices)
R2(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20

! Create DHCP pool
R2(config)# ip dhcp pool mynetwork
R2(dhcp-config)# network 192.168.10.0 255.255.255.0
R2(dhcp-config)# default-router 192.168.10.1
R2(dhcp-config)# dns-server 192.168.10.100
```

✓ Then set client PCs to DHCP — they will get IP from 192.168.10.21 onwards

3.3 DHCP Relay (ip helper-address) — Clients on different subnet

When DHCP server and clients are on DIFFERENT networks — use **ip helper-address** on the router interface facing clients.

```
R1(config)# interface fastEthernet 0/0
R1(config-if)# ip helper-address 10.10.10.2 ← IP of DHCP server
R1(config-if)# exit
R1(config)# router rip
R1(config-router)# network 10.10.10.0
R1(config-router)# network 192.168.30.0
```

■ Relay needs routing (RIP/OSPF/static) between router and DHCP server!

3.4 Full DHCP Lab Configuration

```
! Router 1 (R1) – serial link
R1(config)# int serial 0/0/0 | ip add 10.10.10.1 255.255.255.252 | no shut | clock rate 64000

! Router 2 (R2) – DHCP Server
R2(config)# int serial 0/0/0 | ip add 10.10.10.2 255.255.255.252 | no shut
R2(config)# int fa0/0 | ip add 192.168.10.1 255.255.255.0 | no shut
R2(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.20
R2(config)# ip dhcp pool mynetwork
R2(dhcp-config)# network 192.168.10.0 255.255.255.0
```

```
R2(dhcp-config)# default-router 192.168.10.1
R2(dhcp-config)# dns-server 192.168.10.100
```

4. OSPF — Open Shortest Path First

OSPF is a **link-state** routing protocol — most widely used IGP in large enterprise networks. BGP is the exterior gateway protocol. OSPF uses wildcard masks (opposite of subnet mask).

4.1 OSPF Key Points

- Process ID: 1–65535 (local to router, doesn't need to match neighbors)
- Uses wildcard mask = inverse of subnet mask (e.g. /30 = 0.0.0.3 wildcard)
- Advertises INTERFACES not networks — wildcard determines which interfaces
- Area 0 = backbone area (must exist in all OSPF networks)
- Hello/Dead Interval timers **MUST** match between neighbors

4.2 OSPF Configuration — 3 Router Example

```
! R1 – Interface setup
R1(config)# int serial 0/3 | ip add 10.10.10.1 255.255.255.252 | clock rate 64000 | no shut
R1(config)# int serial 0/2 | ip add 10.10.10.5 255.255.255.252 | clock rate 64000 | no shut

! R2 – Interface setup
R2(config)# int serial 0/2 | ip add 10.10.10.6 255.255.255.252 | no shut
R2(config)# int fa0/0 | ip add 192.168.10.1 255.255.255.0 | no shut

! R3 – Interface setup
R3(config)# int serial 0/2 | ip add 10.10.10.2 255.255.255.252 | no shut
R3(config)# int fa0/0 | ip add 192.168.1.1 255.255.255.0 | no shut

! Enable OSPF on all routers
R1(config)# router ospf 1
R1(config-router)# network 10.10.10.0 0.0.0.3 area 0
R1(config-router)# network 10.10.10.4 0.0.0.3 area 0

R2(config)# router ospf 1
R2(config-router)# network 192.168.10.0 0.0.0.255 area 0
R2(config-router)# network 10.10.10.4 0.0.0.3 area 0

R3(config)# router ospf 1
R3(config-router)# network 192.168.1.0 0.0.0.255 area 0
R3(config-router)# network 10.10.10.0 0.0.0.3 area 0
```

4.3 OSPF Verification & Troubleshooting Commands

Command	Description
show ip protocol	Displays all routing protocol parameters
show ip route	Full IP routing table (O = OSPF routes)
show ip ospf	Basic OSPF process info
show ip ospf interface	OSPF info for all interfaces
show ip ospf interface fa0/0	OSPF info for specific interface
show ip ospf neighbor	Lists all OSPF neighbors and states
show ip ospf neighbor detail	Detailed neighbor info
show ip ospf border-routers	Border and boundary router info
clear ip route *	Clears routing table (forces rebuild)
clear ip ospf process	Resets OSPF, rebuilds neighbors & DB
debug ip ospf events	Shows all OSPF events in real time
debug ip ospf adjacency	Shows DR/BDR election and states
debug ip ospf packets	Shows OSPF packets

4.4 OSPF Timer Commands

```
R(config-if)# ip ospf hello-interval 20 ← Change hello timer (default 10s)
R(config-if)# ip ospf dead-interval 80 ← Change dead timer (default 40s)
```

■ **Hello and Dead interval timers MUST match on both ends for routers to become OSPF neighbors!**

5. RIP — Routing Information Protocol

RIP is a **distance-vector** IGP. Uses **hop count** as metric. Max hops = 15 (16 = unreachable). Broadcasts entire table every **30 seconds**. Works well in small networks.

5.1 RIPv1 vs RIPv2

Feature	RIPv1	RIPv2
Protocol Type	Classful	Classless (supports VLSM)
Updates	Broadcast	Multicast (224.0.0.9)
VLSM Support	No	Yes
Authentication	No	Yes (hashed password)
Triggered Updates	No	Yes (immediate on change)
Max Hops	15	15
Update Interval	30 seconds	30 seconds

5.2 RIP Timers

Timer	Default	Description
Update Timer	30 sec	Interval between routing table broadcasts to all neighbors
Invalid Timer	180 sec	If no update for 180s → route marked invalid, neighbors notified
Holddown Timer	180 sec	Suppresses routing info after route becomes unreachable
Flush Timer	240 sec	Time before invalid route is removed from routing table

5.3 RIP Configuration — Full Example (2 Routers)

```
! R1 (1841Router0)
R1(config)# int fa0/0 | ip add 10.0.0.1 255.0.0.0 | no shut
R1(config)# int fa0/1 | ip add 20.0.0.1 255.0.0.0 | no shut
R1(config)# int serial 0/0/0 | ip add 50.0.0.1 255.0.0.0 | clock rate 64000 | no shut
R1(config)# router rip
R1(config-router)# network 10.0.0.0
R1(config-router)# network 20.0.0.0
R1(config-router)# network 50.0.0.0

! R2 (2811Router1)
R2(config)# int fa0/0 | ip add 30.0.0.1 255.0.0.0 | no shut
R2(config)# int fa0/1 | ip add 40.0.0.1 255.0.0.0 | no shut
R2(config)# int serial 0/0/0 | ip add 50.0.0.2 255.0.0.0 | no shut
R2(config)# router rip
R2(config-router)# network 30.0.0.0
R2(config-router)# network 40.0.0.0
R2(config-router)# network 50.0.0.0
```

5.4 RIP Command Cheat Sheet

Command	Description
router(config)# router rip	Enable RIP routing process
router(config-router)# network w.x.y.z	Advertise directly connected network w.x.y.z
router(config-router)# version 2	Use RIPv2 globally
router(config-router)# version 1	Use RIPv1 only
router(config-router)# no auto-summary	Disable auto summarization (RIPv2)
router(config-router)# no router rip	Disable RIP completely
router(config-router)# passive-interface s0/0/0	Don't send RIP updates out this interface
router(config-router)# no ip split-horizon	Disable split horizon
router(config-router)# timers basic 30 90 180 270 360	Change all RIP timers manually
router# show ip rip database	Show RIP routing database
router# show ip protocols	Show RIP parameters and neighbors
router# show ip route	Show routing table (R = RIP routes)

```
router# debug ip rip
```

Real-time RIP activity

6. Static Routing

Static routing = manually adding routes to routing table. No routing protocol overhead, but must be updated manually when topology changes.

6.1 Advantages & Disadvantages

✓ Advantages	✗ Disadvantages
No CPU overhead on router	Admin must understand entire topology
No bandwidth used between routers	Must manually update all routers when network changes
More secure — admin controls all routes	Not scalable — impractical for large networks

6.2 Static Route Command Syntax

```
Router(config)# ip route [admin_distance] [permanent]
```

OR

```
Router(config)# ip route [admin_distance] [permanent]
```

Parameter	Meaning
destination_network	The network you want to reach
subnet_mask	Subnet mask of the destination network
next_hop_IP	IP address of next router to send packets to
exit_interface	Local interface to send packets out of
admin_distance	Default = 1 (or 0 if exit interface used). Lower = preferred.
permanent	Keep route even if interface goes down (without it: route removed on interface down)

6.3 Static Route — 2 Router Example

Topology: PC0(10.0.0.2) → R1(10.0.0.1 / 20.0.0.1) —serial— R2(20.0.0.2 / 30.0.0.1) → PC1(30.0.0.2)

! R1

```
R1(config)# int fa0/0 | ip add 10.0.0.1 255.0.0.0 | no shut
```

```
R1(config)# int serial 0/0/0 | ip add 20.0.0.1 255.0.0.0 | clock rate 64000 | bandwidth 64 | no shut
```

```
R1(config)# ip route 30.0.0.0 255.0.0.0 20.0.0.2 ← tell R1 how to reach 30.0.0.0 network
```

! R2

```
R2(config)# int fa0/0 | ip add 30.0.0.1 255.0.0.0 | no shut
```

```
R2(config)# int serial 0/0 | ip add 20.0.0.2 255.0.0.0 | no shut
```

```
R2(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1 ← tell R2 how to reach 10.0.0.0 network
```

✓ Test: PC0> ping 30.0.0.2 — should get replies

6.4 Static Route — 4 Router Example

Networks: 10.0.0.0 (R1-PC0) | 20.0.0.0 (R1-R2 serial) | 30.0.0.0 (R2-R3) | 40.0.0.0 (R3-R4 serial) | 50.0.0.0 (R4-PC1)

! R1: needs routes to 30, 40, 50

```
R1(config)# ip route 30.0.0.0 255.0.0.0 20.0.0.2
```

```
R1(config)# ip route 40.0.0.0 255.0.0.0 20.0.0.2
```

```
R1(config)# ip route 50.0.0.0 255.0.0.0 20.0.0.2
```

! R2: needs routes to 10, 40, 50

```
R2(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1
```

```
R2(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.2
```

```
R2(config)# ip route 50.0.0.0 255.0.0.0 30.0.0.2
```

! R3: needs routes to 10, 20, 50

```
R3(config)# ip route 10.0.0.0 255.0.0.0 30.0.0.1
```

```
R3(config)# ip route 20.0.0.0 255.0.0.0 30.0.0.1
```

```
R3(config)# ip route 50.0.0.0 255.0.0.0 40.0.0.2
```

! R4: needs routes to 10, 20, 30

```
R4(config)# ip route 10.0.0.0 255.0.0.0 40.0.0.1
```

```
R4(config)# ip route 20.0.0.0 255.0.0.0 40.0.0.1
```

```
R4(config)# ip route 30.0.0.0 255.0.0.0 40.0.0.1
```

7. Quick Reference — Common Interface & General Commands

Command	Description
Router> enable	Enter privileged EXEC mode
Router# configure terminal	Enter global configuration mode
Router(config)# hostname R1	Set router hostname to R1
R1(config)# interface fa0/0	Enter FastEthernet 0/0 interface config
R1(config)# interface serial 0/0/0	Enter Serial 0/0/0 interface config
R1(config-if)# ip address x.x.x.x mask	Assign IP address to interface
R1(config-if)# no shutdown	Bring interface UP (default is admin down)
R1(config-if)# clock rate 64000	Set clock rate on DCE end of serial cable
R1(config-if)# bandwidth 64	Set bandwidth (in Kbps) on serial interface
R1(config-if)# exit	Return to global config mode
R1(config-if)# end	Return to privileged EXEC mode
R1# show ip route	Show full routing table
R1# show interfaces	Show all interface status
R1# show ip interface brief	Quick overview of all interfaces and IPs
R1# show running-config	Show current running configuration
R1# ping x.x.x.x	Test connectivity to IP address
PC> ipconfig	Show IP config on Cisco Packet Tracer PC
PC> ping x.x.x.x	Test connectivity from PC

7.1 Serial Interface Notes

- Serial cable has two ends: **DTE** and **DCE**. Clock rate is ALWAYS set on the **DCE end**.
- Use **show controllers <interface>** to check which end is DCE.
- For /30 subnets on serial links: wildcard = 0.0.0.3, mask = 255.255.255.252
- WIC-2T module: needed to add serial ports to routers in Packet Tracer (turn off router first, add module, turn on).

7.2 Administrative Distance (AD) — Route Trust Level

Route Type	Admin Distance
Directly Connected	0
Static Route	1
OSPF	110
RIP	120

■ Lower AD = more trusted. If multiple routes to same destination exist, router picks the lowest AD.

EXAM TIPS: (1) Always 'no shutdown' after setting IP (2) Clock rate only on DCE serial end (3) NAT: inside/outside must be on correct interfaces (4) OSPF uses wildcard mask (not subnet mask) (5) RIP max 15 hops (6) DHCP relay = ip helper-address (7) Static route = ip route dest mask next-hop